



VISION

A platform enabling formal recognition of hidden carers' informal caregiving capabilities

CARE-ABLE

System Design Specification

Table of Contents

Chapter 1	5
1.1 Introduction	5
1.2 Problem definition	5
1.3 Project objectives.....	6
1.4 Proposed Solution.....	7
1.5 Scope of the Project.....	8
1.6 Assumptions and Constraints	9
Chapter 2	10
2.1 Software Design Architecture	10
2.2 System Users.....	11
2.3 Functional Requirements	12
2.4 Non-functional Requirements	14
2.5 ER diagrams.....	17
2.6 Data Flow Diagrams	18
2.7 Sequence Diagram	19
2.8 Activity Diagram.....	20
2.9 Interface Design	21
Chapter 3	28
3.1 Implementation	28
3.2 Techniques & Methodology	29
3.3 Security mechanism	31
3.4 System Interface.....	32
3.5 Data tables.....	39
Chapter 4	44
4.1 Testing Plan.....	44
Chapter 5	45
5.1 Reference	45

List of figures

Figure 1 Usecase Diagram	11
Figure 2 ERD Diagram	17
Figure 3 Data Flow Diagrams	18
Figure 4 Sequence Diagram	19
Figure 5 Activity Diagram	20
Figure 6 Wireframe index page	22
Figure 7 fetchers pages	23
Figure 8 About CareAble	24
Figure 9 Register forms	26
Figure 10 Signin view	26
Figure 11 Admin view	27
Figure 12 Index page 1	33
Figure 13 index page 2	33
Figure 14 Index page 3	34
Figure 15 Register Form	34
Figure 16 Register form 2	35
Figure 17 Employee Details	35
Figure 18 Quize 1	36
Figure 19 Quize 2	36
Figure 20 Admin Panel	37
Figure 21 Caregiver Management	37
Figure 22 Employee Management	38
Figure 23 Questions list	38
Figure 24 Certificate	39
Figure 25 Mysql table list	40
Figure 26 Assessment data table	40
Figure 27 Care givers table	41
Figure 28 caregivers details table	41
Figure 29 quiz answers data	42
Figure 30 Employer data	42
Figure 31 Questions table	43
Figure 32 user table	43

List of Tables

Table 1 Test cases.....	43
-------------------------	----

Chapter 1

1.1 Introduction

The care and support sector plays an important role in improving the quality of life of people who are living with disabilities, chronic illnesses, or age-related conditions. Across many communities, a large number of individuals provide daily care to family members or loved ones without receiving formal recognition for their efforts. These individuals are often referred to as “hidden carers” because they do not identify themselves as professional caregivers even though they perform significant caregiving responsibilities. As a result, many hidden carers are excluded from employment opportunities, training pathways, and support services that could help them improve their personal and professional lives.

The CareAble project aims to address this issue by introducing a digital platform that enables hidden carers to assess their caregiving capabilities through structured self-assessment questionnaires. The platform is designed as a mobile-friendly web application that allows users to register, complete skill assessments, generate digital certificates, and access dashboards related to their progress. The assessment system evaluates various caregiving domains such as communication, emotional resilience, practical care, leadership, planning, and digital literacy.

The proposed system focuses on developing a Minimum Viable Product (MVP) that can collect caregiving data and provide formal recognition for informal caregiving skills. By aligning caregiving abilities with recognised skill categories in the Australian Skills Classification and the Care and Support Economy Strategy, the project helps carers gain visibility and confidence in their abilities. The CareAble platform therefore serves as both a recognition tool and a stepping stone toward future employment and professional development opportunities within the care industry.

1.2 Problem definition

Many individuals who provide care for elderly family members, people with disabilities, or individuals with chronic illnesses perform essential caregiving tasks every day without formal recognition or professional support. These hidden carers often develop valuable practical,

emotional, and organizational skills through experience, yet their capabilities remain undocumented and unrecognized. Because they do not possess official qualifications or certifications, they may struggle to access employment opportunities or career pathways within the healthcare and support sectors.

Another major issue is the lack of a dedicated digital system that allows hidden carers to evaluate and demonstrate their caregiving abilities. Existing systems mainly focus on professional healthcare workers and do not provide a simple mechanism for informal carers to assess their skills. As a result, carers often feel isolated, unsupported, and disconnected from the formal workforce. Additionally, organizations and employers have limited ways to verify the practical caregiving competencies of individuals who have gained experience informally rather than through academic training.

The absence of a standardized self-assessment and certification platform also creates difficulties in collecting reliable caregiving data for research and workforce planning. Without proper data collection and skill evaluation, policymakers and industry organizations cannot fully understand the contributions and needs of hidden carers. Therefore, there is a strong need for a secure, accessible, and scalable platform that can assess caregiving skills, generate recognized digital certificates, and support the long-term development of hidden carers within the care economy.

1.3 Project objectives

The main objective of the CareAble project is to develop a user-friendly web-based platform that enables hidden carers to formally assess and recognise their caregiving capabilities. The system is designed to provide carers with a structured and accessible method for evaluating their knowledge, practical skills, and emotional strengths across multiple caregiving domains.

The specific objectives of the project are as follows:

- To design and develop a mobile-friendly web application that supports multiple user roles, including carers and administrators.

- To create a secure user authentication and registration system that allows users to create accounts and manage their profiles safely.
- To develop a structured self-assessment questionnaire system based on caregiving skill categories such as communication, emotional resilience, practical care, planning, leadership, and digital literacy.
- To implement an automated scoring mechanism that evaluates assessment responses and categorises capability levels into strength areas, growth areas, and support areas.
- To generate digital certificates automatically upon completion of assessments in order to provide formal recognition of informal caregiving skills.
- To provide administrators with dashboard and analytics features for viewing assessment data, managing users, and validating certificates.
- To create a scalable and extensible software architecture that can support future enhancements, additional user roles, and expanded caregiving assessments.
- To support the collection of caregiving data that may contribute to future research and improvements within the care and support industry.

1.4 Proposed Solution

The proposed solution for this project is the development of the CareAble platform, a modern web-based application designed to recognise and support hidden carers through digital self-assessment and certification. The platform will provide carers with an opportunity to evaluate their caregiving skills in a structured and professional manner while also helping organisations better understand the capabilities of informal caregivers.

The system will include a responsive landing page containing information about the platform, its objectives, partner organisations, and contact details. Users will be able to register and log in securely using email and password authentication. Different user roles such as carers, administrators, and employers will be supported within the system architecture to ensure flexibility for future expansion.

After registration, carers will be able to complete skill assessments consisting of multiple questionnaire domains based on Likert-scale responses. These assessments will measure caregiving competencies including communication skills, emotional resilience, self-care management, digital literacy, cultural awareness, and leadership abilities. The system will automatically calculate scores for each domain and classify the results into capability categories such as “Strength Area,” “Growth Area,” and “Support Area.” Once the assessment is completed, the platform will generate a downloadable PDF certificate containing the carer’s details, assessment results, completion date, and top capability areas. This certificate will serve as evidence of the carer’s informal caregiving experience and skills. Administrators will also have access to dashboards for monitoring user activities, viewing analytics, managing assessments, and validating certificate authenticity. The proposed solution will be developed using modern web technologies with a focus on scalability, usability, and low-cost cloud deployment. The system architecture will be designed in a modular and extensible manner so that future developers can easily add new features, user roles, and assessment modules without major system redesign. Ultimately, the CareAble platform aims to bridge the gap between informal caregiving experience and formal recognition within the healthcare and support industry.

1.5 Scope of the Project

CareAble is a web-based platform designed to manage the end-to-end interaction between caregivers and employers.

Core Capabilities:

- **User Management:** Separate registration and login portals for Caregivers and Employers.
- **Profile Building:** Caregivers can create comprehensive digital resumes including:
 - Technical skills and specialized care expertise.
 - Professional work history and experience levels.
 - Educational background and certifications.
 - Personal bio-data and contact information.
- **Automated Certification:** Upon successful profile completion and verification, the system generates and issues an official "careAble Certification" delivered via email.

- **Employer Search & Discovery:** A robust search engine for employers to filter caregivers based on specific requirements (skills, location, or education).
- **Job Offer Workflow:** Direct communication tools allowing employers to extend formal job offers within the platform.

Out of Scope:

- **In-Platform Payments:** Initial versions may not include integrated payroll or escrow services.
- **Background Checks:** While the system validates educational levels, deep criminal background checks may be handled externally.

1.6 Assumptions and Constraints

The system is designed based on several assumptions regarding users, technology, and operating conditions. It is assumed that all users, including caregivers and employers, have access to a stable internet connection and a device such as a smartphone, tablet, or computer to access the web application.

It is also assumed that users will provide accurate and truthful information when registering and creating their profiles. The effectiveness of the caregiver matching and assessment process depends heavily on the quality and correctness of the data entered into the system.

Another assumption is that caregivers possess basic digital literacy, enabling them to complete online assessments and manage their profiles without significant difficulty. Similarly, employers are expected to have the ability to navigate the system, search for candidates, and make hiring decisions using the platform.

The system assumes that assessments are designed appropriately to evaluate caregiver skills and that the scoring logic accurately reflects the competency level of each caregiver. It is also assumed that administrators will regularly maintain and update assessment content and system data.

In addition, it is assumed that the system will be deployed in a secure environment with proper hosting and database support, ensuring availability and data protection.

Chapter 2

2.1 Software Design Architecture

The CareAble system follows a client-server architecture designed to provide a scalable, secure, and efficient platform for caregiver management and assessment processes. The frontend of the system is developed using React.js, which provides a responsive and interactive user interface for caregivers, employers, and administrators. React enables component-based development, improving maintainability, reusability, and overall user experience across different devices and screen sizes. The frontend communicates with the backend through RESTful APIs using HTTP requests and responses.

The backend is implemented using Node.js with the Express.js framework, which handles the core business logic and server-side operations of the application. The backend processes user requests, manages authentication and authorization, validates data, handles assessment and onboarding functionalities, and communicates with the database. Express.js provides a lightweight and flexible framework for developing scalable APIs and supports middleware integration for security and request handling.

Communication between the frontend and backend is achieved through REST API architecture, where data is exchanged in JSON (JavaScript Object Notation) format. JSON provides a lightweight and platform-independent method for transmitting data efficiently between the client and server. The RESTful API structure ensures modularity, maintainability, and easier integration with third-party systems or future mobile applications.

The system also follows a three-tier architecture consisting of the Presentation Layer, Business Logic Layer, and Data Layer. The Presentation Layer represents the React frontend interface that users interact with directly. The Business Logic Layer is managed by the Node.js and Express backend, where all application rules, validations, authentication mechanisms, and workflow processing are handled. The Data Layer consists of the database system, which stores caregiver details, employer information, assessments, quiz results, onboarding data, and

certificates. Database connectivity is established through secure server-side database queries and API integration, ensuring reliable data management and consistency.

2.2 System Users

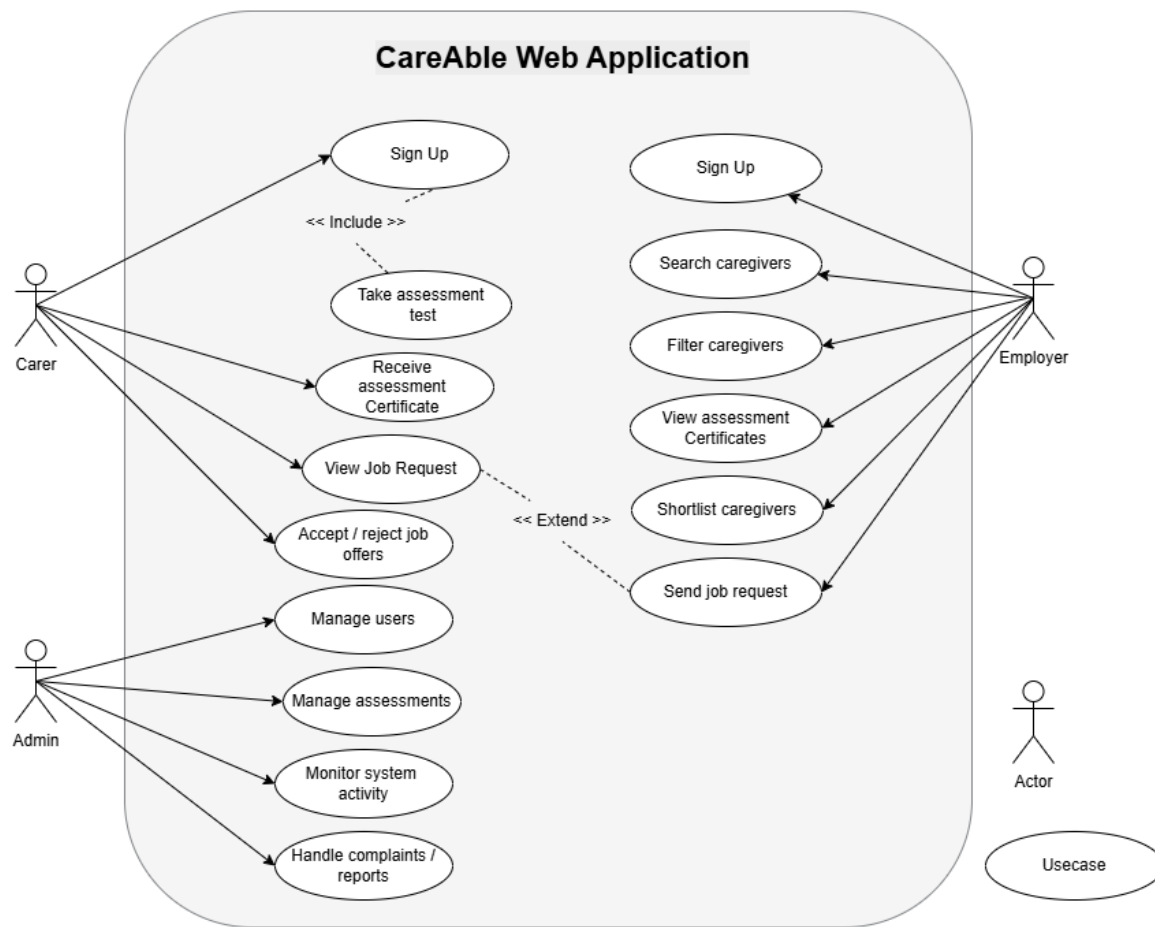


Figure 1 Usecase Diagram

Main Users of the System

Caregiver

Person who provides care services

Wants to get hired

Employer / Client

Person or organization looking for a caregiver

Admin

Manages the system and users

Use Case Scenarios for Caregivers

Sign Up: A facility that lets new caregivers sign up by offering necessary information about themselves.

Login: A feature that facilitates login of registered caregivers into their dashboards.

Take Assessment: This is a working part of the application through which caregivers can take assessment tests or evaluate themselves.

View Result: Caregivers will be able to view their results and even download certificates generated by the system.

2.3 Functional Requirements

The careAble system will perform the following primary functions to facilitate the connection between caregivers and employers:

➤ **User Registration and Authentication**

- The system shall allow users to create accounts using email and password authentication.
- The system shall support multiple user roles such as Carer, Admin, and Employer.
- The system shall validate user credentials during login.
- The system shall allow users to reset forgotten passwords securely.

- The system shall collect onboarding information including name, email, phone number, postcode, and consent details during registration.

➤ **Landing Page**

- The system shall provide a responsive landing page accessible to all visitors.
- The landing page shall display information about the platform, its objectives, and partner organisations.
- The landing page shall include a contact form for user inquiries.
- The landing page shall provide navigation links for login and registration.

➤ **Carer Skill Assessment**

- The system shall allow carers to access and complete multiple skill assessments.
- The system shall provide questionnaires using a Likert scale from 1 to 5.
- The system shall group assessment questions into caregiving domains such as communication, emotional resilience, practical care, and leadership.
- The system shall allow users to save incomplete assessments and resume them later.
- The system shall allow carers to retake assessments multiple times.

➤ **Scoring and Evaluation**

- The system shall automatically calculate scores for each assessment domain.
- The system shall classify results into:
 - Strength Area (4.0 – 5.0)
 - Growth Area (3.0 – 3.9)
 - Support Area (1.0 – 2.9)
- The system shall display assessment results to the user after completion.

➤ **Certificate Generation**

- The system shall generate a PDF certificate automatically after assessment completion.
- The certificate shall contain:
 - Certificate ID
 - Carer name

- Email address
- Completion date
- Completed domains
- Top capability areas
- The system shall allow users to download their certificates.

➤ **Dashboard and Data Management**

- The system shall provide a user dashboard for carers to view completed assessments and certificates.
- The system shall provide an admin dashboard for monitoring platform activities.
- The system shall allow admins to search and filter users.
- The system shall allow admins to edit, remove, or add assessment questions and domains.
- The system shall display analytical reports and visual charts related to assessment results.

➤ **Certificate Validation**

- The system shall allow admins and employers to verify the authenticity of certificates using certificate IDs.

➤ **Role Management**

- The system shall allow admins to assign and remove user roles.
- The system shall support future extension for additional user roles.

2.4 Non-functional Requirements

Non-functional requirements describe the quality standards, system constraints, and performance expectations of the CareAble platform.

➤ **Usability**

- The system shall provide a simple, modern, and user-friendly interface.
- The platform shall be mobile-friendly and responsive across different screen sizes.
- The system shall be easy to navigate for users with limited technical knowledge.

➤ **Performance**

- The system shall load pages within an acceptable response time under normal usage conditions.
- The platform shall support up to 10,000 users efficiently.
- The system shall process assessment submissions and certificate generation without significant delay.

➤ **Security**

- The system shall securely store user passwords using encryption techniques.
- The system shall protect personal user information from unauthorised access.
- The system shall provide secure authentication and session management.
- The system shall ensure that only authorised admins can access sensitive data and management features.

➤ **Scalability**

- The software architecture shall support future expansion and additional features without major redesign.
- The system shall support the addition of new assessments, domains, and user roles.

➤ **Reliability**

- The platform shall maintain stable operation with minimal downtime.
- The system shall ensure data redundancy and backup mechanisms to prevent data loss.
- The system shall accurately save and retrieve assessment results and certificates.

➤ **Maintainability**

- The system shall use modular and well-structured code for easier maintenance.
- The application shall support future updates and bug fixes efficiently.
- The database design shall support easy modification and extension.

➤ **Compatibility**

- The system shall function properly on major web browsers such as Chrome, Firefox, Edge, and Safari.

- The system shall support both desktop and mobile devices.

➤ **Deployment**

- The system shall support cloud deployment using low-cost infrastructure services such as AWS free-tier solutions.
- The deployment process shall use Infrastructure as Code practices for easier management.

➤ **Availability**

- The platform shall be accessible online 24/7 for users.
- The system shall ensure continuous access to completed assessments and certificates.

➤ **Data Privacy and Compliance**

- The system shall obtain user consent before collecting personal information.
- The system shall support anonymised data usage for research purposes.
- The platform shall comply with privacy and ethical data handling standards.

2.5 ER diagrams

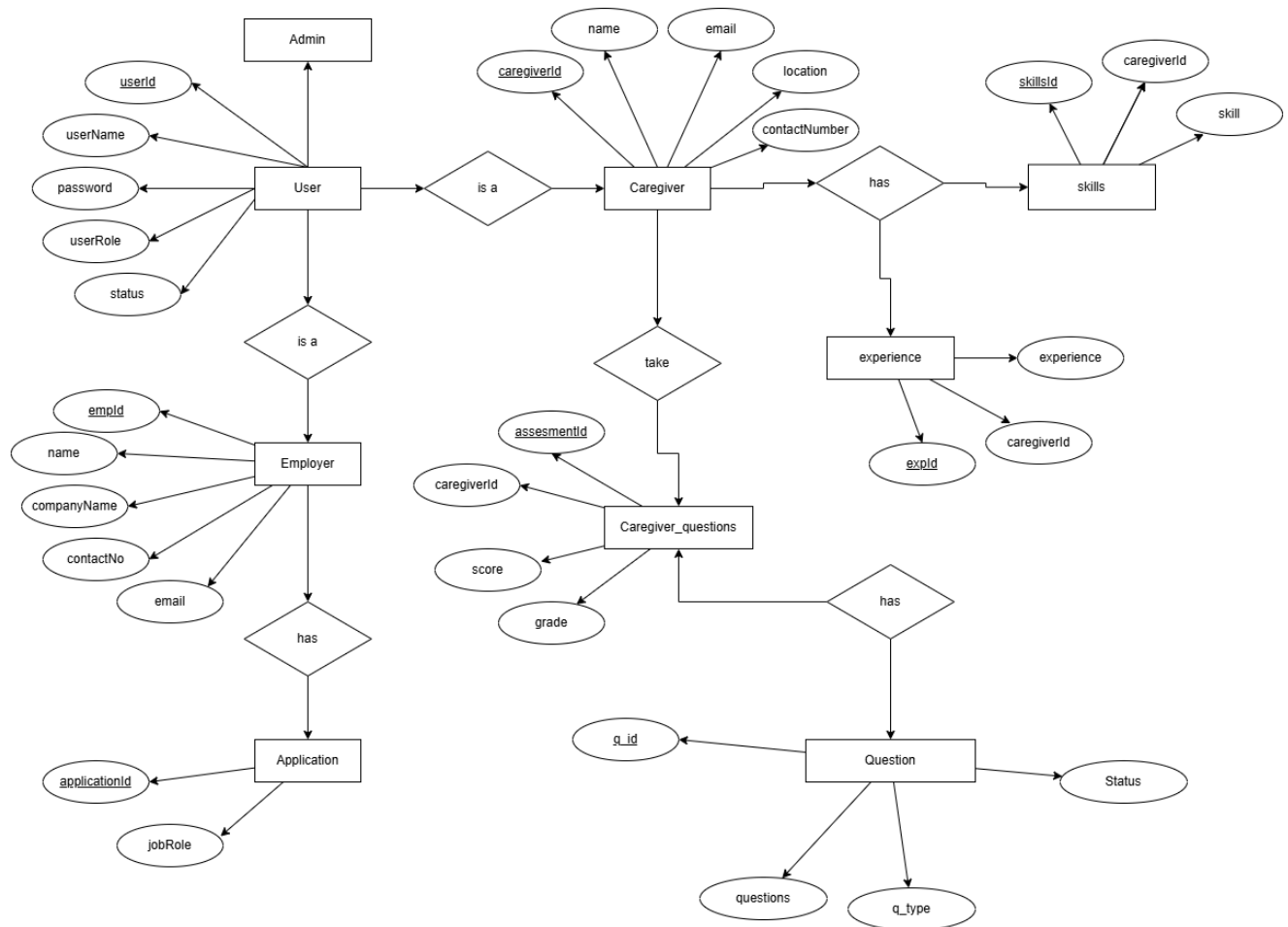


Figure 2 ERD Diagram

2.6 Data Flow Diagrams

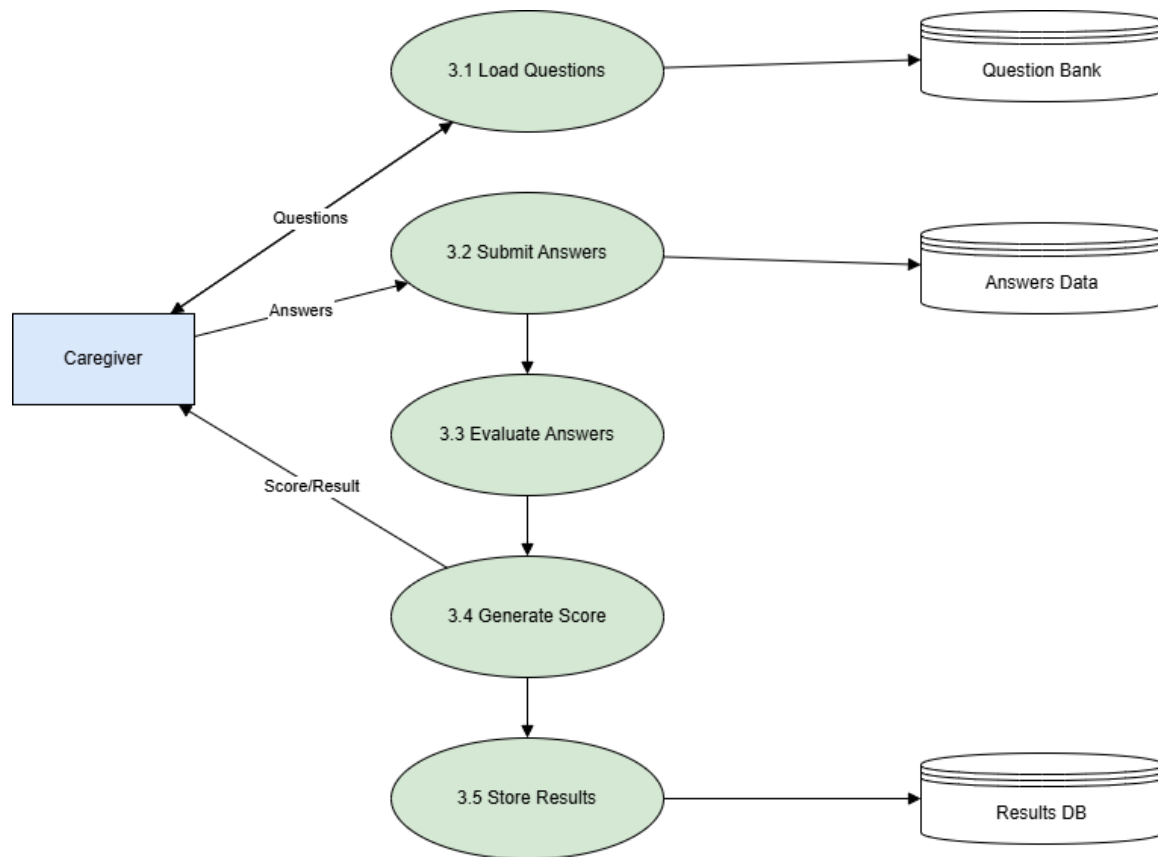


Figure 3 Data Flow Diagrams

2.7 Sequence Diagram

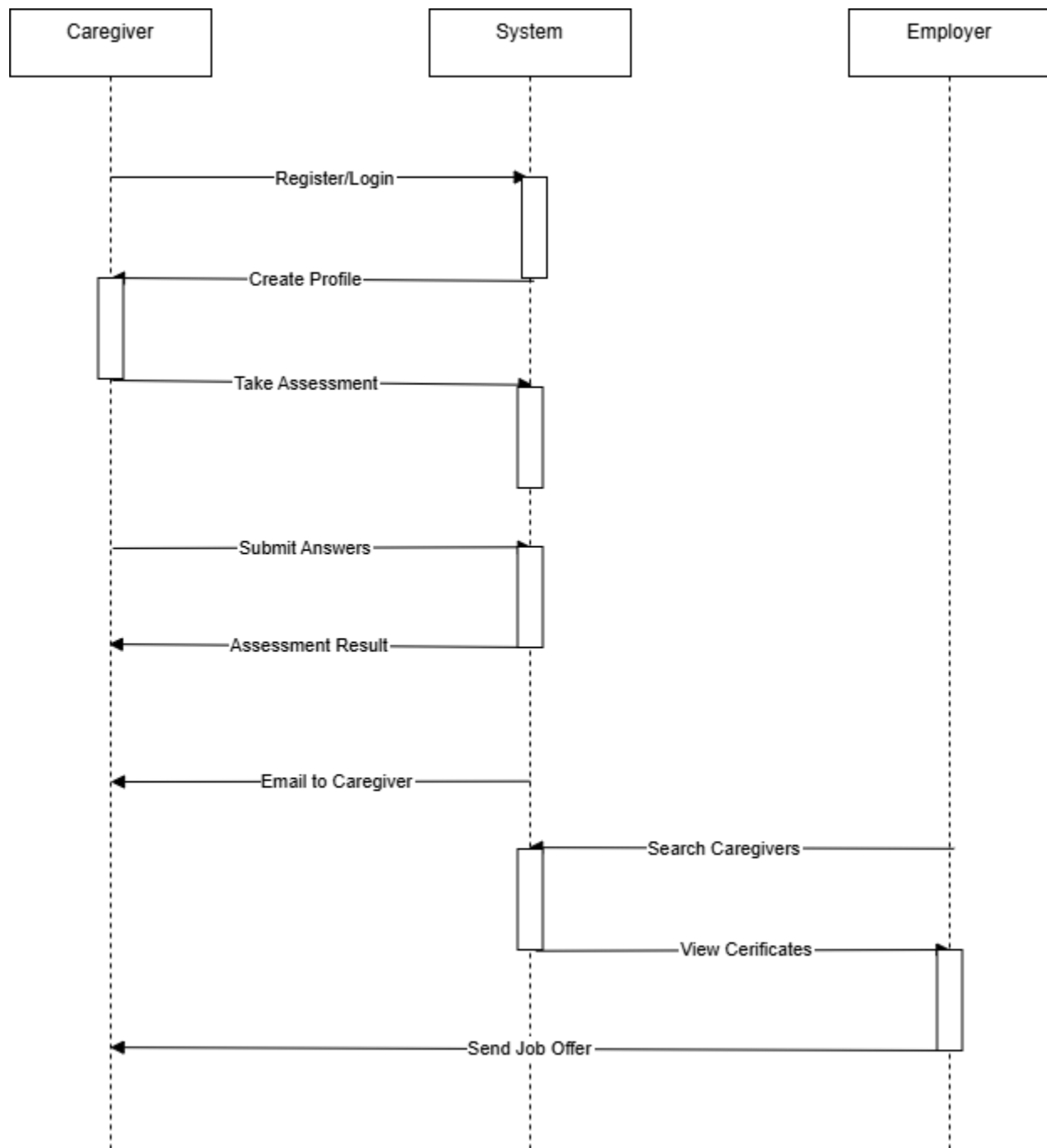


Figure 4 Sequence Diagram

2.8 Activity Diagram

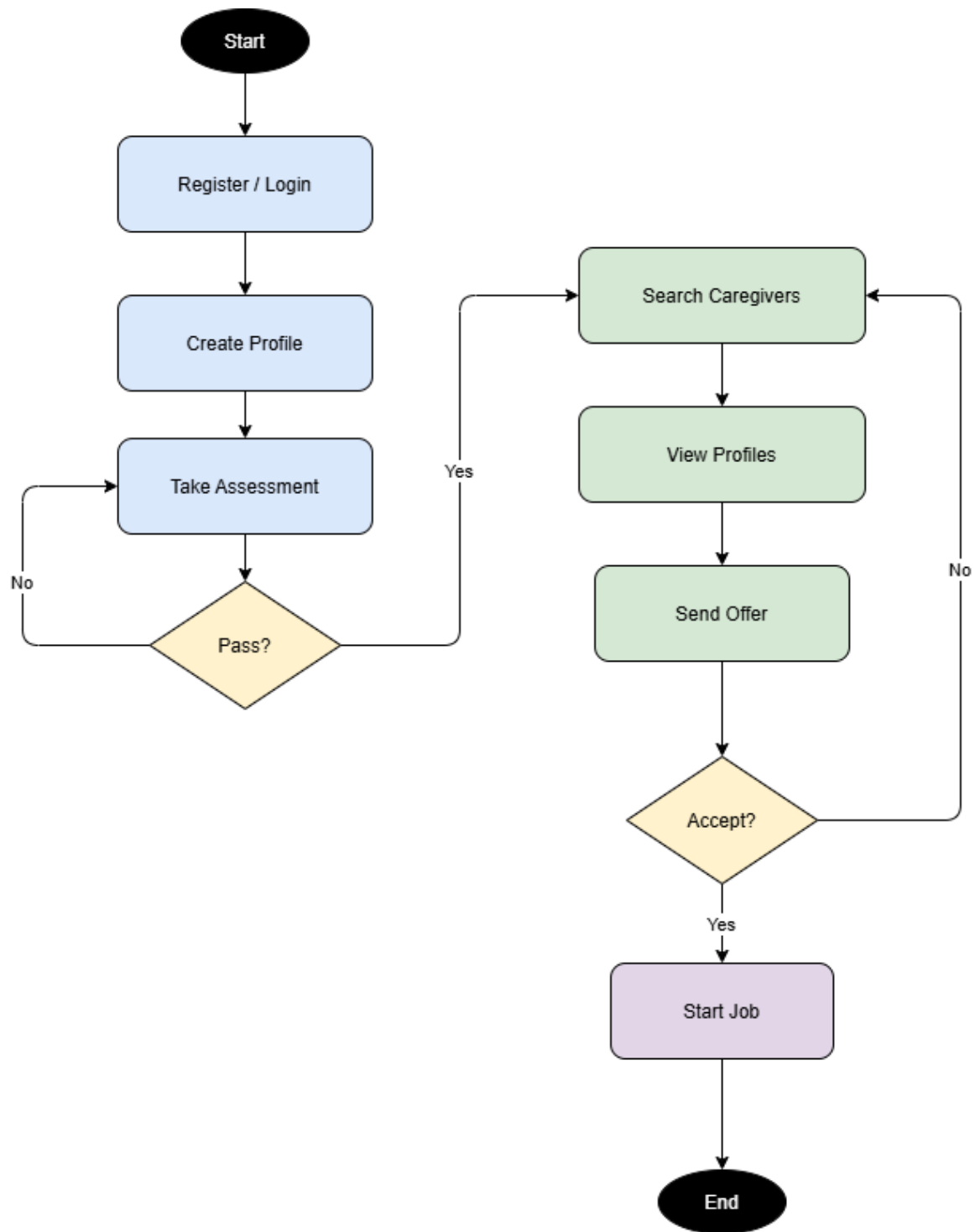


Figure 5 Activity Diagram

2.9 Interface Design

The interface design section of the CareAble project demonstrates the visual structure and user interaction flow of the proposed system. This section includes both the initial prototype designs created using Figma and the final implemented user interfaces developed during the system implementation phase. The purpose of the interface design is to ensure that the platform provides a simple, modern, user-friendly, and responsive experience for all users, especially carers who may have limited technical knowledge.

The Figma designs were used during the planning and design stage to visualise the layout, navigation structure, colour themes, and user interactions before development began. These prototype screens helped identify user requirements, improve usability, and ensure consistency across different pages of the system. Key interfaces designed in Figma include the landing page, registration and login pages, skill assessment pages, dashboards, certificate generation screens, and admin management panels.

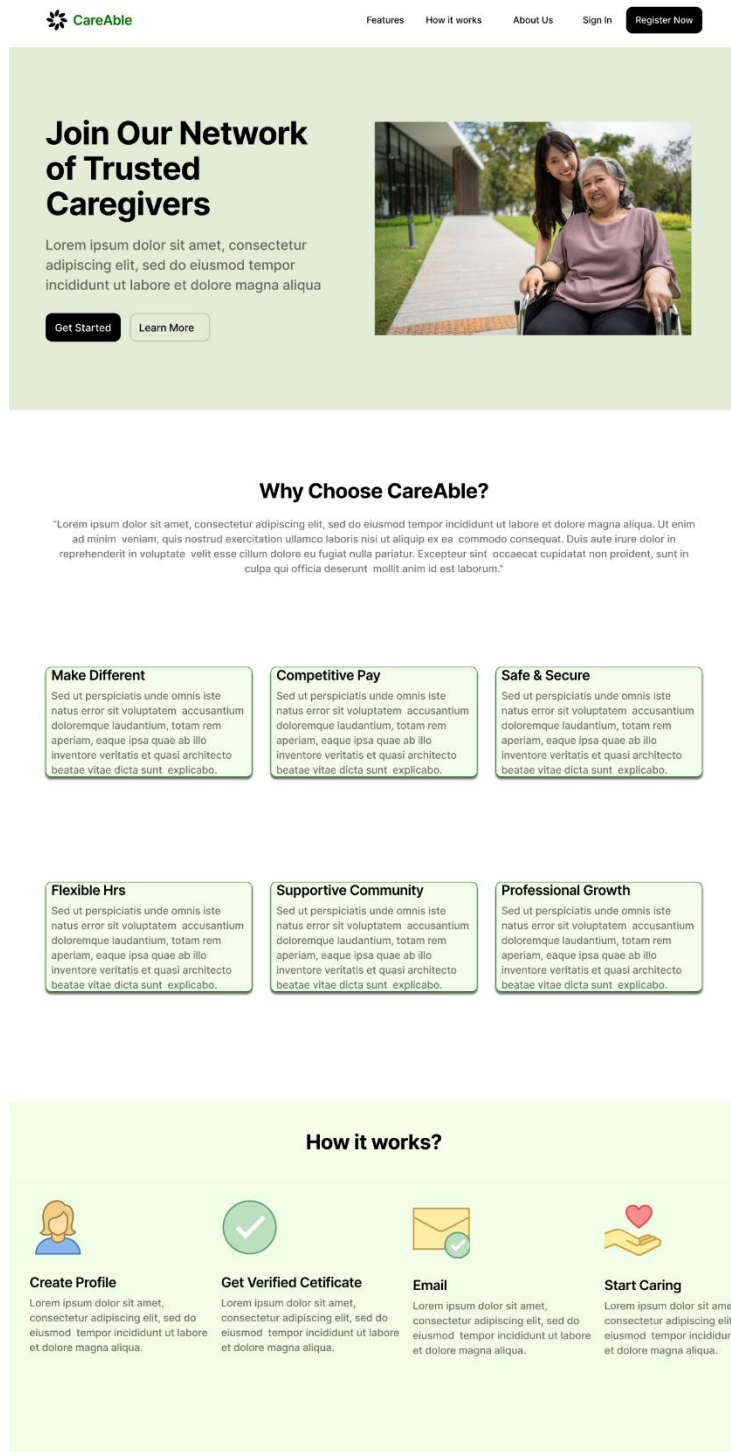
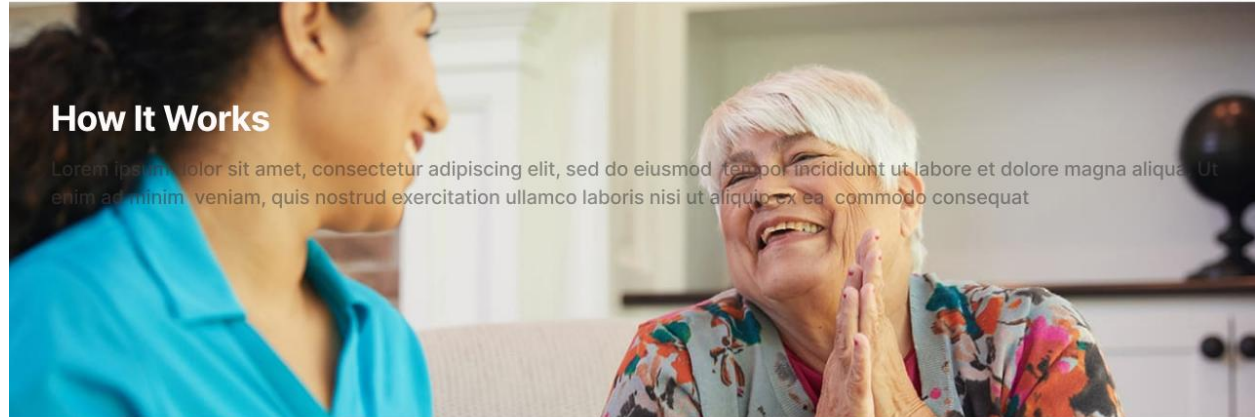


Figure 6 Wireframe index page



How It Works

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat

How it works?



Create Profile

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



Get Verified Certificate

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



Email

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Start Caring

Lorem ipsum dolor sit amet, consectetur adipiscing elit, eiusmod tempor incididunt et dolore magna aliqua.

CareAble

Descriptive line about what your company does.



Features

Core features
Pro experience
Integrations

Learn more

Blog
Case studies
Customer stories
Best practices

Support

Contact
Support
Legal

Figure 7 fetchers pages

About Us



Introduction

CareAble is a web-based caregiver assessment and certification system designed to support the evaluation, management, and recognition of caregivers in a structured and efficient way. The platform aims to bridge the gap between caregivers and employers by providing a reliable system to assess skills, track performance, and generate verified digital certificates.

Achievements

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat



Completes assessments

100+

Receives certification

100+

Reviews

1000+

CareAble

Descriptive line about what your company does.



Features

Core features
Pro experience
Integrations

Learn more

Blog
Case studies
Customer stories
Best practices

Support

Contact
Support
Legal

Figure 8 About CareAble

Register With Us

Already Registered?

Call out a feature, benefit, or value of your site, then link to a page where people can learn more about it.

[Sign In](#)

Personal Information

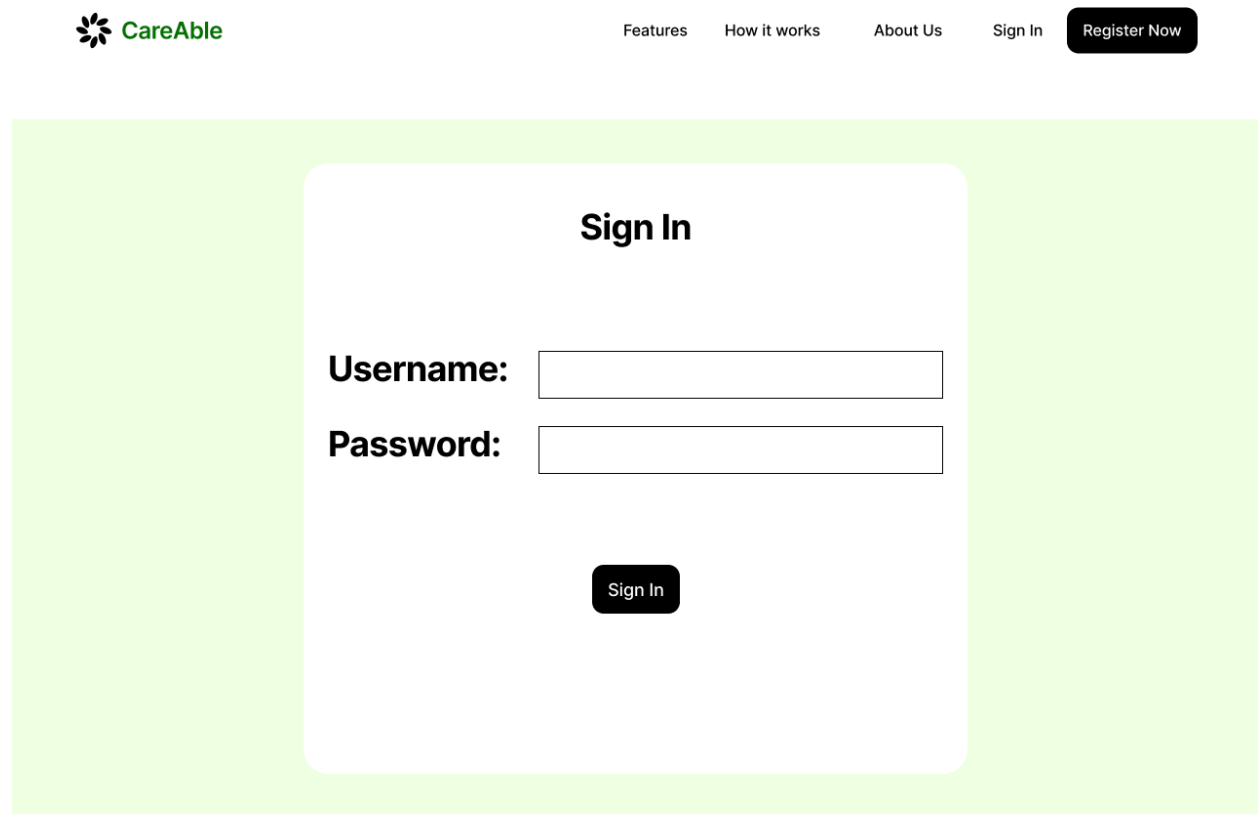
Full Name:	
Date of Birth:	YYYY/MM/DD
Gender:	☐ Male ☐ Female
National ID / Passport Number:	
Profile Photo:	Upload

Contact Details

Email Address:	
Home Address:	
City:	
Mobile Number:	

[Next](#)

Figure 9 Register forms



The image shows the top section of the CareAble website. At the top left is the CareAble logo, which consists of a green flower-like icon followed by the text "CareAble". To the right of the logo is a navigation menu with the following links: "Features", "How it works", "About Us", "Sign In", and "Register Now". The "Register Now" link is highlighted with a dark background. Below the navigation menu is a large light green rectangular area. In the center of this area is a white rounded rectangle containing the "Sign In" form. The form has the title "Sign In" at the top, followed by two input fields labeled "Username:" and "Password:". Below these fields is a dark button labeled "Sign In".

CareAble

Features How it works About Us Sign In Register Now

Sign In

Username:

Password:

Sign In

Figure 10 Signin view

Dashboard

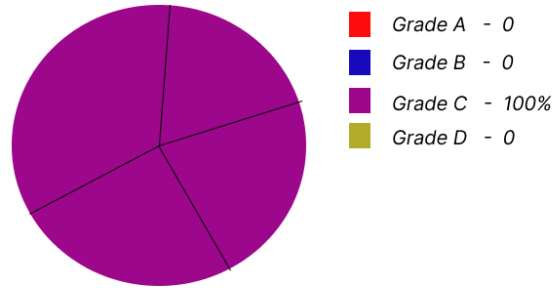
New Caregivers

1. Saige Fuentes
2. Bowen Higgins
3. Leighton Kramer
4. Kylan Gentry
5. Amelie Griffith
6. Franklin Sierra
7. Marceline Avila
8. Jaylen Blackwell
9. Saoirse Conrad
10. Dilan Wolf

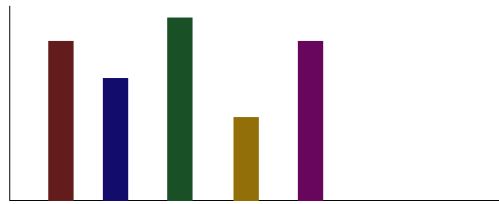
New Employers

1. Christian Jay Abarca
2. John Michael Alcaide

Certificates for this month



Monthly Assigners



CareAble.Au

Figure 11 Admin view

Chapter 3

3.1 Implementation

The CareAble system was implemented as a modern web-based application designed to provide a simple and accessible platform for hidden carers to assess and recognise their caregiving capabilities. The development process focused on building a responsive, scalable, and user-friendly solution that satisfies both the functional and non-functional requirements identified during the system analysis phase.

The frontend of the application was developed using React.js to create a dynamic and interactive user interface. React components were used to build key system modules such as the landing page, user authentication pages, dashboards, skill assessment forms, and certificate viewing pages. Tailwind CSS was used to improve the visual appearance and responsiveness of the interface, ensuring compatibility across desktop and mobile devices.

The backend of the system was implemented using Node.js and Express.js. The backend handles business logic, user authentication, assessment processing, certificate generation, and communication with the database. RESTful APIs were developed to manage data exchange between the frontend and backend systems efficiently.

MySQL DB was used as the primary database management system to store user accounts, assessment questions, skill assessment results, certificate details, and administrative data. The database structure was designed in a flexible and extensible manner to support future system enhancements and additional user roles.

Several major modules were implemented within the CareAble platform, including:

- User registration and authentication module
- Multi-role access management system
- Skill assessment management module
- Automated scoring and evaluation system
- PDF certificate generation module

- Admin dashboard and analytics module
- Certificate validation system

The application was developed using a modular architecture to improve maintainability and scalability. Git and GitHub were used for version control and collaboration during the development process. The final system was tested across multiple browsers and devices to ensure consistent functionality and responsiveness.

3.2 Techniques & Methodology

The development of the CareAble platform followed a structured software engineering methodology to ensure effective planning, implementation, testing, and deployment of the system. The Agile development methodology was selected because it supports iterative development, continuous feedback, and flexibility when handling changing project requirements.

The project was divided into several development phases including requirement gathering, system analysis, interface design, implementation, testing, and deployment. During each iteration, specific system features were developed, reviewed, and improved based on testing outcomes and stakeholder feedback.

Several software development techniques were applied throughout the project:

Requirement Analysis

System requirements were gathered from the project brief and analysed to identify functional and non-functional requirements. User roles, workflows, and system features were carefully documented before development began.

UI/UX Prototyping

Figma was used to design wireframes and user interface prototypes before implementation. This helped visualise the system structure, improve navigation flow, and ensure consistency across the application.

Component-Based Development

React.js component-based architecture was used to improve code reusability and maintainability. Reusable components such as navigation bars, forms, dashboards, and tables reduced development complexity and improved consistency.

RESTful API Development

REST APIs were developed using Express.js to handle communication between the frontend and backend systems. APIs were used for user authentication, assessment submissions, certificate generation, and data retrieval.

Database Design

MySQL DB collections and schemas were designed to support flexible data storage and efficient data retrieval. Relationships between users, assessments, certificates, and roles were carefully structured to maintain data consistency.

Testing Techniques

Several testing techniques were applied during development, including:

- Unit Testing
- Integration Testing
- User Interface Testing
- Functional Testing
- Cross-browser Testing

Testing helped identify bugs, validate system behaviour, and improve overall reliability and usability.

Responsive Web Design

Responsive design techniques were used to ensure the system functions correctly on desktops, tablets, and mobile devices. Flexible layouts and adaptive components improved accessibility for different users.

Overall, the selected methodologies and techniques helped ensure that the CareAble platform was developed efficiently while maintaining high levels of usability, scalability, and maintainability.

3.3 Security mechanism

Security was considered an important aspect during the development of the CareAble platform because the system handles sensitive personal information and assessment data related to carers. Several security mechanisms were implemented to protect user data, maintain privacy, and prevent unauthorised access to the system.

User Authentication and Authorization

The system uses secure email and password authentication to verify user identities during login. Role-based access control mechanisms were implemented to ensure that users can only access features relevant to their assigned roles. For example, carers can complete assessments and view certificates, while administrators can manage users and assessment data.

Password Protection

User passwords are encrypted before being stored in the database using hashing algorithms such as bcrypt. This prevents passwords from being stored as plain text and reduces the risk of credential theft.

Secure API Communication

The backend APIs were designed to validate requests and prevent unauthorised access to sensitive system resources. Authentication tokens were used to maintain secure user sessions and protect restricted endpoints.

Input Validation and Data Sanitisation

Input validation techniques were implemented on both frontend and backend systems to prevent invalid or malicious data submissions. User inputs such as forms, login credentials, and assessment responses are validated before processing.

Certificate Validation Mechanism

Each generated certificate includes a unique certificate ID that can be verified through the system. This mechanism helps prevent certificate duplication and ensures the authenticity of assessment results.

Data Privacy and Confidentiality

The platform collects personal data only with user consent. Sensitive user information is stored securely within the database and access is restricted to authorised users only. Research-related data can be anonymised to protect user identity.

Session Management

Secure session handling techniques were implemented to reduce the risk of session hijacking and unauthorised access. Users are required to log in again after session expiration.

3.4 System Interface

After finalising the prototype designs, the actual system interfaces were developed using modern frontend technologies. The implemented interfaces closely follow the original Figma designs while introducing additional improvements for responsiveness, accessibility, and performance. The actual interfaces demonstrate how users interact with the fully functional CareAble platform, including completing assessments, viewing results, downloading certificates, and managing user data.

Index Page

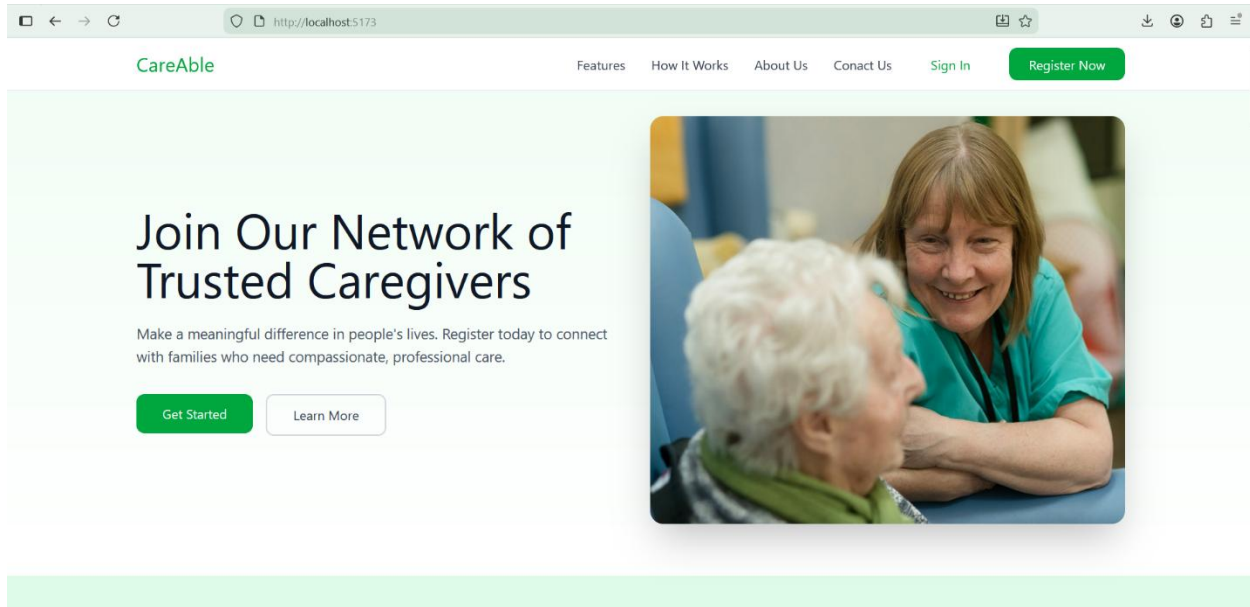


Figure 12 Index page 1

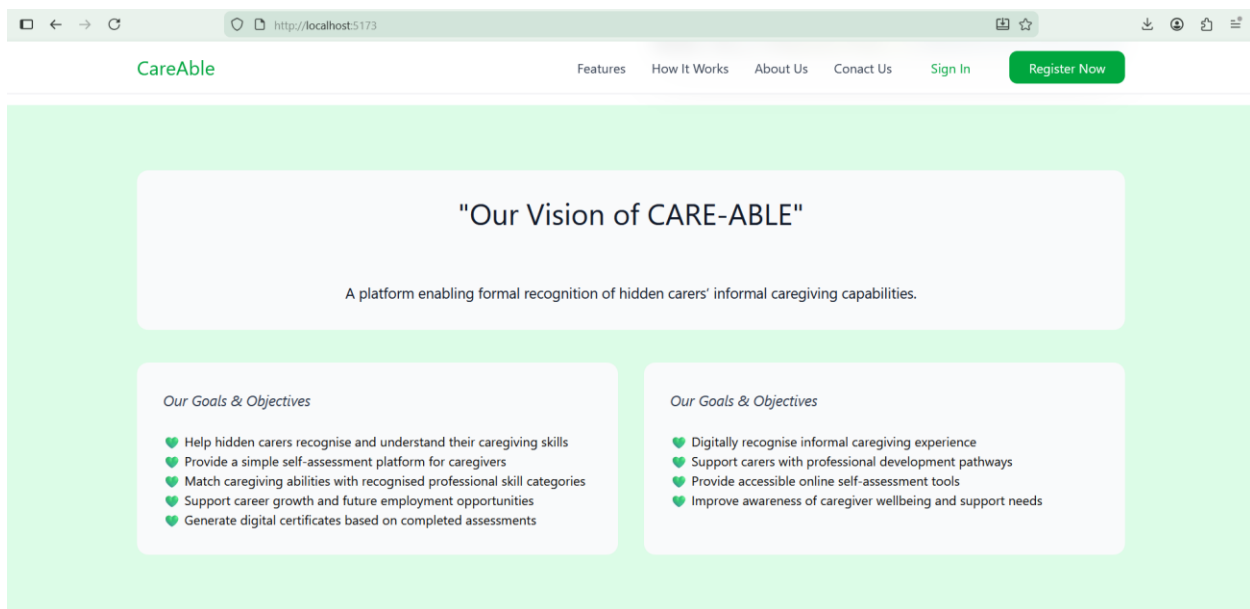


Figure 13 index page 2

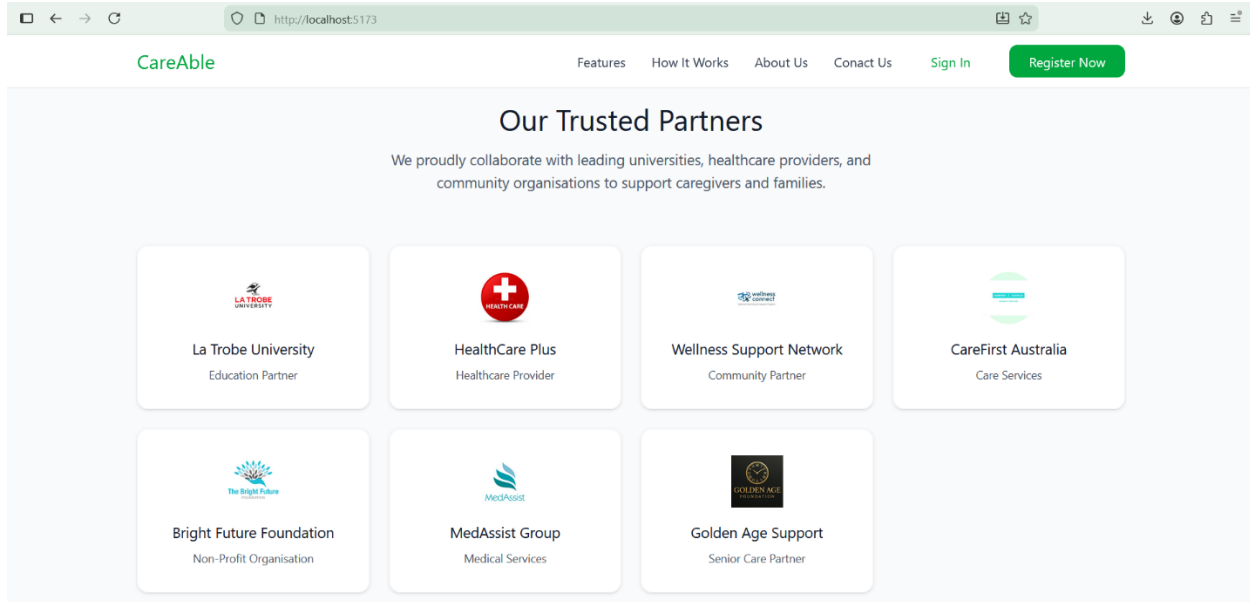


Figure 14 Index page 3

Figure 15 Register Form

CareAble

Features How It Works About Us Contact Us Sign In Register Now

Address
Address

Postcode
Postcode

☐ Accept terms of service and privacy policy.

☐ Consent to have data anonymised and used for research purposes.

Account Setup

Password

Confirm Password

Register & go to Login

Figure 16 Register form 2

CareAble

Features How It Works About Us Contact Us Sign In Register Now

Hidden Worker Status

Are you working at the moment?
Select

Are you looking for work?
Select

Have you applied for jobs within 4 weeks?
Select

Industry Interest

CALD Status

Speak language other than English?
Select

Which language?
Select

Caregiving Information

How did you hear about this app?
Text input

What brings you here?
Text input

Who do you care for?
Text input

Age of person you care for
Select

Figure 17 Employee Details

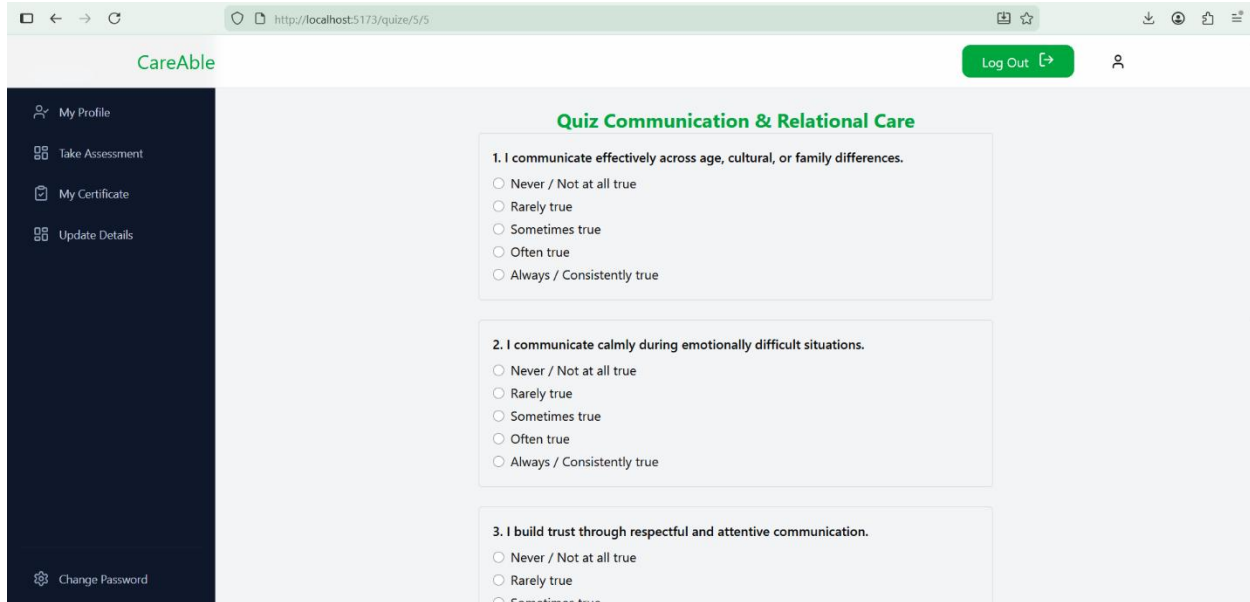


Figure 18 Quiz 1

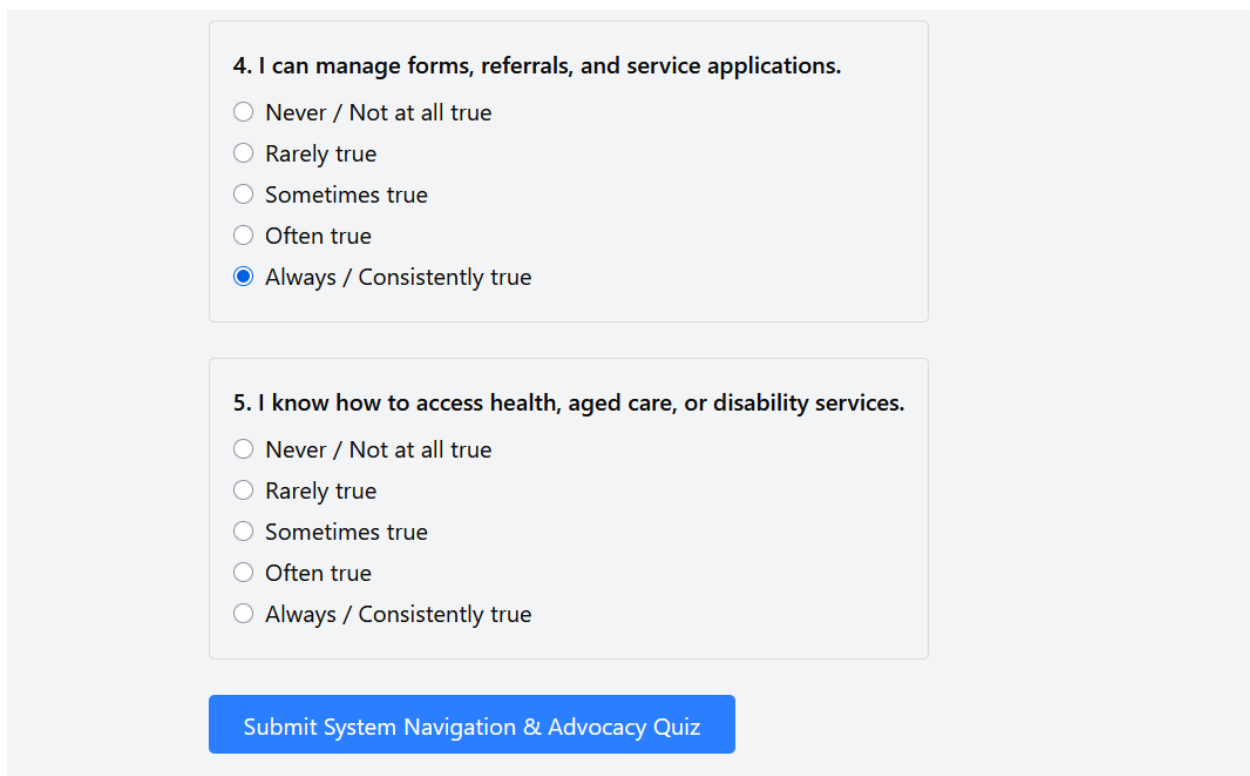


Figure 19 Quiz 2

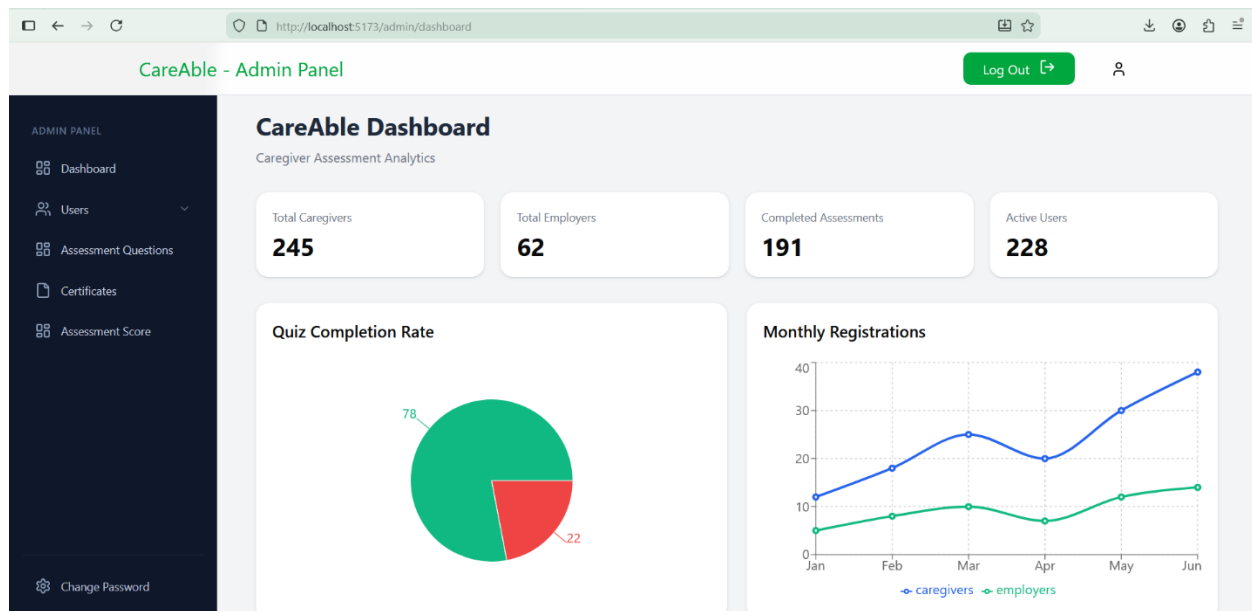


Figure 20 Admin Panel

CareAble - Admin Panel

Caregiver Management

Log Out

ADMIN PANEL

- Dashboard
- Users
- Care Givers
- Employers
- Assessment Questions
- Certificates
- Assessment Score
- Change Password

Refresh

Search Caregiver... Search Filter by Postcode All Status Filter

ID	Name	Mobile	Email	Postcode	Status	Actions
1	Anne Smith	0712588996	anne@g.c	444444	ACT	Edit Delete View
2	Lucy Fearnly	0715558896	lucy@g.c	555555	ACT	Edit Delete View
3	Mark Vega	0712400223	mark@g.c	2101	ACT	Edit Delete View
4	Griffin Siyran	0712400553	griff@gmail.com	2328	ACT	Edit Delete View
5	Miyasri Kangana	0715558896	miyas@gmail.com	2321	ACT	Edit Delete View

Showing 1 - 5 of 5

Previous 1 Next

Figure 21 Caregiver Management

CareAble - Admin Panel Log Out Refresh

ADMIN PANEL

- Dashboard
- Users
- Care Givers
- Employers
- Assessment Questions
- Certificates
- Assessment Score
- Change Password

Employer Management

Search employer... Search Filter by city All Status Filter

ID	Name	Mobile	Email	Postcode	Status	Actions
1	Jack Ryan	433111222	jack.r@gmail.com	2000	ACT	Edit Delete
2	Ruby Matthews	455444333	ruby.m@gmail.com	3000	ACT	Edit Delete
3	Cooper Evans	422777888	coop.e@gmail.com	4000	ACT	Edit Delete
4	Sienna Morris	466999000	sienna.m@gmail.com	5000	ACT	Edit Delete
5	Eli Watson	411888555	eli.w@gmail.com	6000	ACT	Edit Delete

Showing 1 - 5 of 12 Previous 1 2 3 Next

Figure 22 Employee Management

All Questions			
ID	Domains	Question	Actions
60	Leadership & Coordination	I take initiative when leadership is required.	Edit Delete
59	Leadership & Coordination	I keep others informed about care needs.	Edit Delete
58	Leadership & Coordination	I manage conflict constructively.	Edit Delete
57	Leadership & Coordination	I delegate responsibilities appropriately.	Edit Delete
56	Leadership & Coordination	I coordinate caregiving tasks among others.	Edit Delete
55	Planning & Organisation	I stay organised under pressure.	Edit Delete
54	Planning & Organisation	I adjust plans when unexpected situations arise.	Edit Delete

Figure 23 Questions list

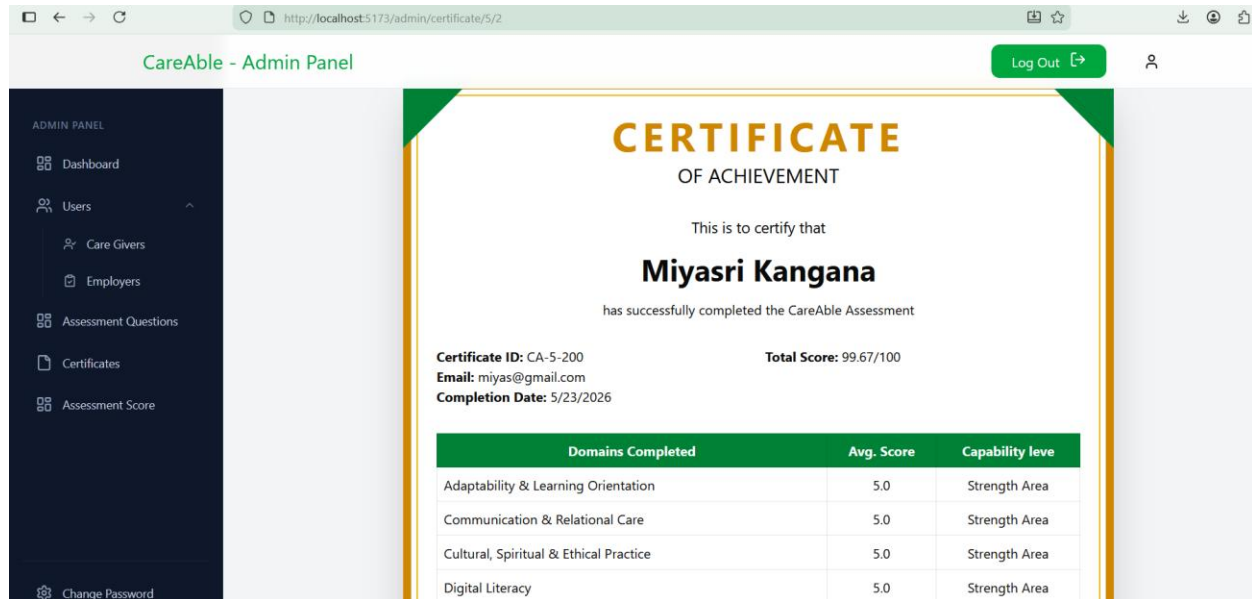


Figure 24 Certificate

3.5 Data tables

The CareAble system uses a MySQL relational database to store and manage system data efficiently. The database consists of multiple interconnected tables designed to support caregiver management, employer operations, assessments, onboarding processes, and certificate generation. Each table contains specific attributes with primary and foreign key relationships to maintain data consistency and integrity. The database structure enables secure data storage, efficient querying, and reliable communication between the backend application and the data layer. The following tables are used within the CareAble system to manage user information, assessments, onboarding details, quiz responses, and administrative functionalities.

Server: 127.0.0.1 » Database: careable

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking Designer

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ admin	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ application	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ assessments	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ caregivers	★ Browse Structure Search Insert Empty Drop	35	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ caregiver_care_types	★ Browse Structure Search Insert Empty Drop	21	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ caregiver_onboarding	★ Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ caregiver_questions	★ Browse Structure Search Insert Empty Drop	360	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ caregiver_skills	★ Browse Structure Search Insert Empty Drop	19	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ care_types	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ employer	★ Browse Structure Search Insert Empty Drop	12	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ questions	★ Browse Structure Search Insert Empty Drop	60	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ skills	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ user	★ Browse Structure Search Insert Empty Drop	50	InnoDB	utf8mb4_general_ci	16.0 KiB	-
13 table(s)	Sum	576	InnoDB	utf8mb4_general_ci	304 KiB	0 B

⬆ ☐ Check all With selected:

Figure 25 Mysql table list

Server: 127.0.0.1 » Database: careable » Table: assessments

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

✓ Showing rows 0 - 5 (6 total, Query took 0.0008 seconds.)

SELECT * FROM `assessments`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

	assessmentid	caregiverid	score	grade	attempt	added_at
☐ Edit Copy Delete	3	1	267.00	NULL	1	NULL
☐ Edit Copy Delete	4	2	299.00	NULL	1	NULL
☐ Edit Copy Delete	5	3	267.00		1	0000-00-00 00:00:00
☐ Edit Copy Delete	6	4	299.00		1	0000-00-00 00:00:00
☐ Edit Copy Delete	7	5	267.00		1	0000-00-00 00:00:00
☐ Edit Copy Delete	8	5	299.00		2	0000-00-00 00:00:00

Figure 26 Assessment data table

Server: 127.0.0.1 » Database: careable » Table: caregivers

Showing rows 0 - 24 (35 total, Query took 0.0042 seconds)

SELECT * FROM `caregivers`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Restore column order Number of rows: 25 Filter rows: Search this table Sort by key: None

		id	first_name	last_name	dob	email	mobile	address	gender	nic_passport	photo	city	password
☐	Edit Copy Delete	1	Anne	Smith	2000-05-14	anne@g.c	0712588996	22A, Perth	NULL	NULL	NULL	444444	\$2b\$10\$pjW1jPrZkPBHRcagqmKR
☐	Edit Copy Delete	2	Lucy	Fearnly	1998-07-01	lucy@g.c	0715558896	55, Flower lane, Melbon	NULL	NULL	NULL	555555	\$2b\$10\$wsQZGdPgxyD89sKNRKj
☐	Edit Copy Delete	3	Mark	Vega	1995-03-20	mark@g.c	0712400223	95 Queen Street, Narrabeen, New South Wales	NULL	NULL	NULL	2101	\$2b\$10\$Ywky/cmX.rGJV3HwBTct
☐	Edit Copy Delete	4	Griffin	Syran	1992-05-07	griff@gmail.com	0712400553	43 Woordens Road, Yarrawa, New South Wales	NULL	NULL	NULL	2328	\$2b\$10\$3pXIT06L.v9F64xdlKqK
☐	Edit Copy Delete	5	Miyasri	Kangana	1983-08-05	miyas@gmail.com	0715558896	80 Gaggin Street, Morpeth, New South Wales	NULL	NULL	NULL	2321	\$2b\$10\$ZYbomfoOeXFzULED6hN
☐	Edit Copy Delete	7	Oliver	Smith	1990-03-14	oliver.s@gmail.com	412345678	12 Flinders Street, Melbourne, Victoria	NULL	NULL	NULL	3000	\$2b\$10\$pjW1jPrZkPBHRcagqmKR
☐	Edit Copy Delete	8	Charlotte	Brown	1988-11-22	char.b@gmail.com	415987654	45 George Street, Sydney, New South Wales	NULL	NULL	NULL	2000	\$2b\$10\$pjW1jPrZkPBHRcagqmKR

Figure 27 Care givers table

Server: 127.0.0.1 » Database: careable » Table: caregiver_onboarding

Showing rows 0 - 4 (5 total, Query took 0.0006 seconds)

SELECT * FROM `caregiver\_onboarding`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

		onboarding_id	caregiverId	current_work_status	looking_for_work	applied_jobs_4weeks	industry_interest	speak_other_language	other_language	heard_about
☐	Edit Copy Delete	1	1	No	N	Y	Child Care	Y	Cantonese	Internet
☐	Edit Copy Delete	2	2	Yes FT	Y	Y	Health care	Y	Arabic	Friend
☐	Edit Copy Delete	3	5	Yes FT	Y	Y	Child Care	Y	Spanish	Internet
☐	Edit Copy Delete	4	3	Yes FT	Y	Y	Health care	Y	Arabic	Friend
☐	Edit Copy Delete	5	4	Yes FT	Y	Y	Health care	Y	Arabic	Friend

Figure 28 caregivers details table

Server: 127.0.0.1 » Database: careable » Table: caregiver_questions

Showing rows 0 - 24 (360 total, Query took 0.0011 seconds.)

SELECT * FROM `caregiver\_questions`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | ☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

			qu_id	caregiverId	attempt	q_id	answer
☐	Edit	Copy	Delete	1	1	1	5 5
☐	Edit	Copy	Delete	2	1	1	4 5
☐	Edit	Copy	Delete	3	1	1	3 5
☐	Edit	Copy	Delete	4	1	1	2 5
☐	Edit	Copy	Delete	5	1	1	1 2
☐	Edit	Copy	Delete	6	1	1	10 5
☐	Edit	Copy	Delete	7	1	1	9 3
☐	Edit	Copy	Delete	8	1	1	8 4
☐	Edit	Copy	Delete	9	1	1	7 3
☐	Edit	Copy	Delete	10	1	1	6 4
☐	Edit	Copy	Delete	11	1	1	15 3
☐	Edit	Copy	Delete	12	1	1	14 5
☐	Edit	Copy	Delete	13	1	1	13 3
☐	Edit	Copy	Delete	14	1	1	12 3
☐	Edit	Copy	Delete	15	1	1	11 5

Console

Figure 29 quiz answers data

Server: 127.0.0.1 » Database: careable » Table: employer

Showing rows 0 - 11 (12 total, Query took 0.0008 seconds.)

SELECT * FROM `employer`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

			empld	first_name	last_name	mobile	email	status	dob	companyName	city	password
☐	Edit	Copy	Delete	1	Jack	Ryan	433111222	jack.r@gmail.com	ACT	1991-04-18	2000	\$2b\$10\$ppjW1jPrZkPBHRcagqmKRQuBPozEW9Wif.WBxcGIX
☐	Edit	Copy	Delete	2	Ruby	Matthews	455444333	ruby.m@gmail.com	ACT	1995-11-02	3000	\$2b\$10\$ppjW1jPrZkPBHRcagqmKRQuBPozEW9Wif.WBxcGIX
☐	Edit	Copy	Delete	3	Cooper	Evans	422777888	coop.e@gmail.com	ACT	1987-08-27	4000	\$2b\$10\$ppjW1jPrZkPBHRcagqmKRQuBPozEW9Wif.WBxcGIX
☐	Edit	Copy	Delete	4	Sienna	Morris	466999000	sienna.m@gmail.com	ACT	1993-01-15	5000	\$2b\$10\$ppjW1jPrZkPBHRcagqmKRQuBPozEW9Wif.WBxcGIX

Figure 30 Employer data

Server: 127.0.0.1 » Database: careable » Table: questions

Figure 31 Questions table

Server: 127.0.0.1 » Database: careable » Table: user

Showing rows 0 - 24 (50 total, Query took 0.0011 seconds)

SELECT * FROM `user`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

+ Options

				userid	userName	password	userRole	status
☐	Edit	Copy	Delete	1	admin	\$2b\$10\$Mtlji.QahUOmIKQT5w2IOeY1RVLxYXJbUELb4y99IS...	ADMIN	1
☐	Edit	Copy	Delete	2	nadee3@g.c	\$2b\$10\$Mtlji.QahUOmIKQT5w2IOeY1RVLxYXJbUELb4y99IS...	CAREGIVER	1
☐	Edit	Copy	Delete	3	ban@gmail.com	\$2b\$10\$JstYoBpQ4uMLFDj8FElaV.Yfwho1T470BacEbsjCyby...	EMPLOYER	1
☐	Edit	Copy	Delete	4	anne@g.c	\$2b\$10\$piW1jPrZkPBHRcagqmKRquBPozEW9Wif.WBxcGIX0aR...	CAREGIVER	1
☐	Edit	Copy	Delete	5	lucy@g.c	\$2b\$10\$wsQZGdPgxyD89sKNRKYzseAs6TSmkARvdTCm8fRbj...	CAREGIVER	1
☐	Edit	Copy	Delete	6	mark@g.c	\$2b\$10\$Ywky/cmX.rGJV3HwBTctudPRPP9VHTaMJAOGDS8ycE...	CAREGIVER	1
☐	Edit	Copy	Delete	7	griff@gmail.com	\$2b\$10\$3pXiT06Llv9F64xdlKqK.OFIsJfz.pTBnUasDqfOVn...	CAREGIVER	1
☐	Edit	Copy	Delete	8	miyas@gmail.com	\$2b\$10\$ZYbomfoOeXFzULED6hNIWleir.gtAOd5tmIQ1K.11ABZ...	CAREGIVER	1
☐	Edit	Copy	Delete	9	oliver.s@gmail.com	\$2b\$10\$piW1jPrZkPBHRcagqmKRquBPozEW9Wif.WBxcGIX0aR...	CAREGIVER	1
☐	Edit	Copy	Delete	10	char.b@gmail.com	\$2b\$10\$piW1jPrZkPBHRcagqmKRquBPozEW9Wif.WBxcGIX0aR...	CAREGIVER	1
☐	Edit	Copy	Delete	11	will.j@gmail.com	\$2b\$10\$piW1jPrZkPBHRcagqmKRquBPozEW9Wif.WBxcGIX0aR...	CAREGIVER	1
☐	Edit	Copy	Delete	12	amelia.m@gmail.com	\$2b\$10\$piW1jPrZkPBHRcagqmKRquBPozEW9Wif.WBxcGIX0aR...	CAREGIVER	1
☐	Edit	Copy	Delete	13	lucas.d@gmail.com	\$2b\$10\$piW1jPrZkPBHRcagqmKRquBPozEW9Wif.WBxcGIX0aR...	CAREGIVER	1
☐	Edit	Copy	Delete	14	mia.w@gmail.com	\$2b\$10\$piW1jPrZkPBHRcagqmKRquBPozEW9Wif.WBxcGIX0aR...	CAREGIVER	1

Figure 32 user table

Chapter 4

4.1 Testing Plan

We have created a test plan for testing.

Test Case ID	Test Scenario	Steps	Expected Result
CA_TC_01	User Registration – Valid Data	Enter valid name, email, password, phone → Submit form	User account created successfully and redirected to dashboard/login
CA_TC_02	User Registration – Invalid Email	Enter email as user@com → Submit form	Validation error shown: “Invalid email format”
CA_TC_03	Login – Valid Credentials	Enter correct email and password → Click login	User successfully logged in and redirected to dashboard
CA_TC_04	Login – Invalid Password	Enter correct email + wrong password → Submit	Error message: “Incorrect password”
CA_TC_05	Assessment Submission – Valid	Answer all questions → Submit assessment	Answers saved and result/score generated
CA_TC_06	Assessment – Missing Answers	Leave some questions blank → Submit	Warning message: “Please answer all questions”
CA_TC_07	Certificate Generation	Complete assessment with passing score → Open certificate page	Certificate generated and available to view/download
CA_TC_08	Frontend links	Load home page → Open Links	Open the paths
CA_TC_09	Frontend hash links	Load About page → Open hash Links in home page	Open home page and go to correct div
CA_TC_10	Unauthorized Access Control	Logout → Access /admin/users via URL	Redirect to login page or “Access Denied” message

Chapter 5

5.1 Reference

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